1. State two properties of graphite

2. State two properties of diamond

3. What is an allotrope?

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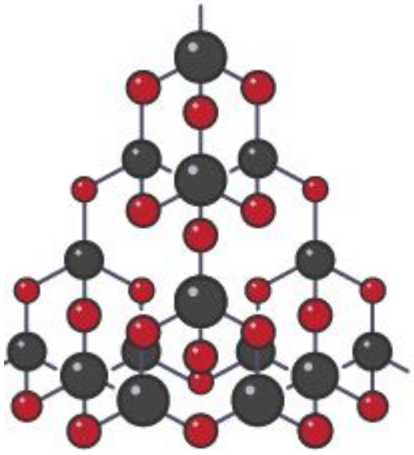
DO NOW

Giant covalent structures

1. Diamond
2. Graphite
3. Silicon dioxide- main compound found in sand.

- It is a giant covalent structure. It contains many silicon and oxygen atoms.
- All the atoms are linked to each other by strong covalent



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ACTIVATION

Metallic bonding

Learning objective

1. Describe that metals form giant structures.
2. Explain how metal ions are held together.
3. Explain delocalisation of electrons

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Keywords

- Delocalised electrons
- Metal ions
- Metallic bonding
- ‘Sea’ of electrons

Metallic bonding

Metal bonded to another metal, the outer electron can move throughout the metal (delocalised).

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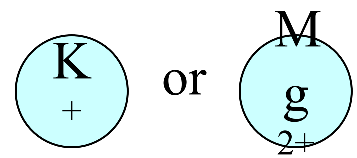
ACTIVATION

Metallic Bonding

In ionic bonding we know metals lose electrons to form +ve ions

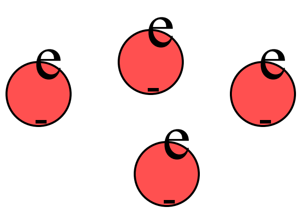


M is any metal

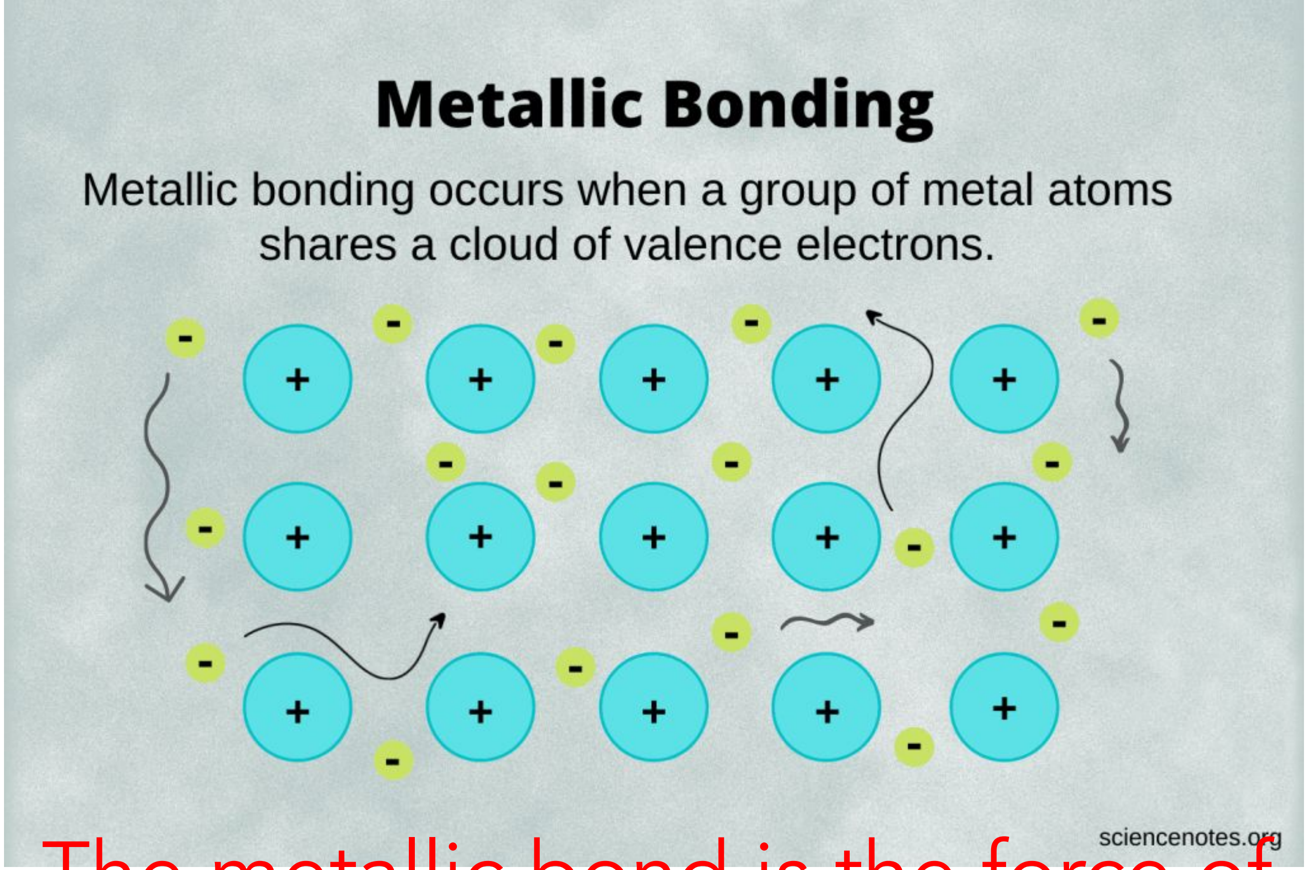
This is to achieve a completely full outer electron shell

This also happens in metallic bonding

- but now the electrons have nowhere to go!



The electrons form a "sea of -ve charge" that flows around the +ve metal ions.
The -ves attract the +ves and vice versa



- The metallic bond is the force of attraction between these free-moving (delocalised) electrons and positive metal ions.

Why do the metal ions form a regular shape?

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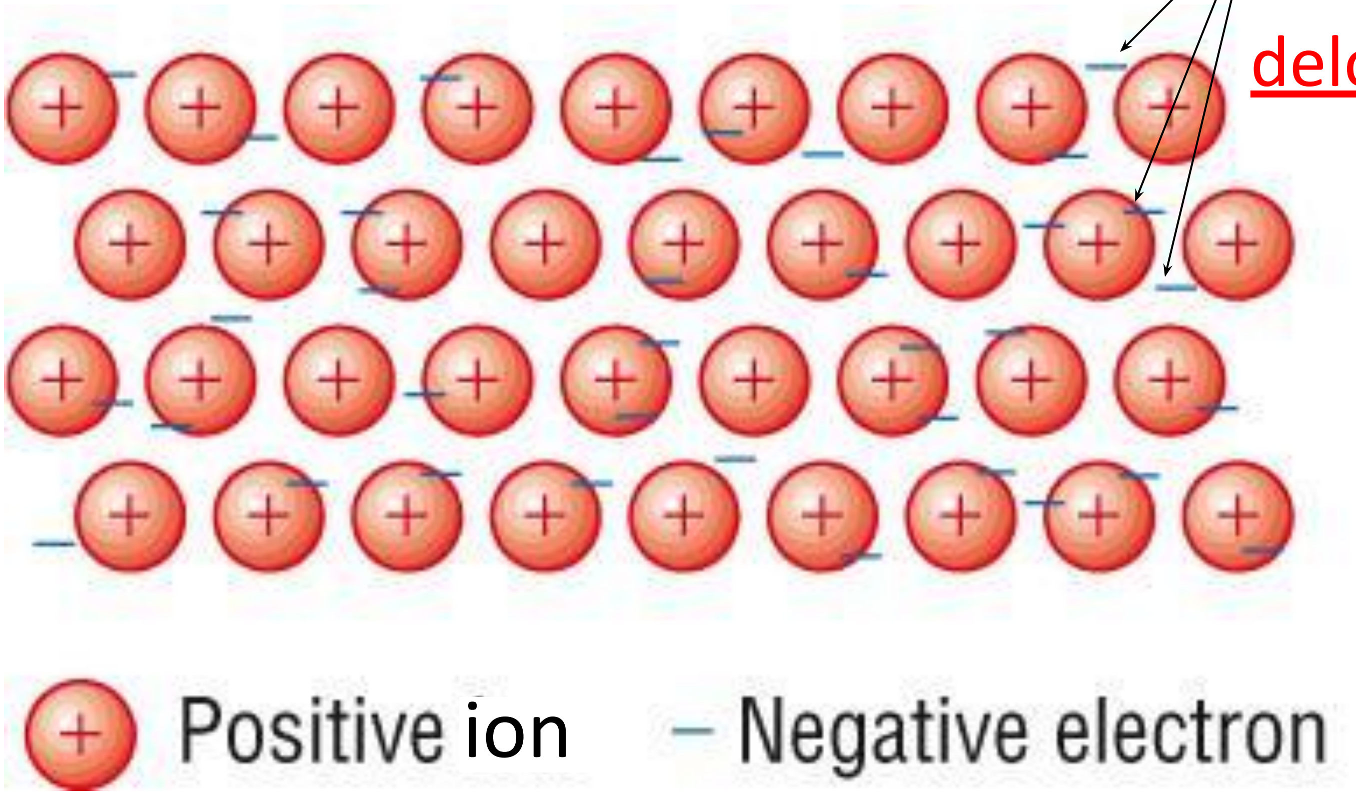
ACTIVATION

How are Metals Bonded?

Electrons can move <u>delocalised</u>

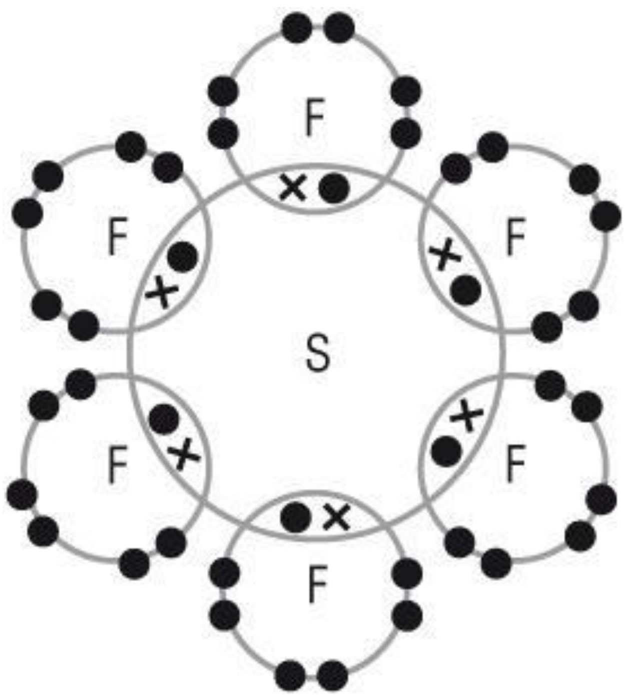


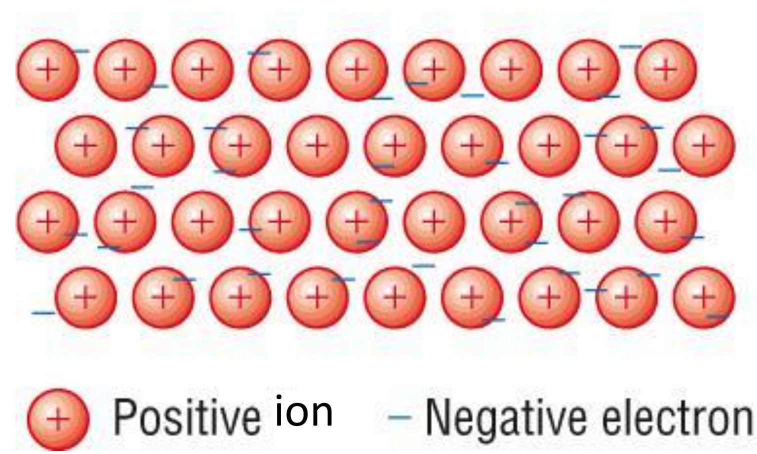
Why do the metal ions form a regular shape?

Definition- Re-cap

- Metallic bonding is the strong electrostatic attraction between positive metal ion and delocalised electrons.

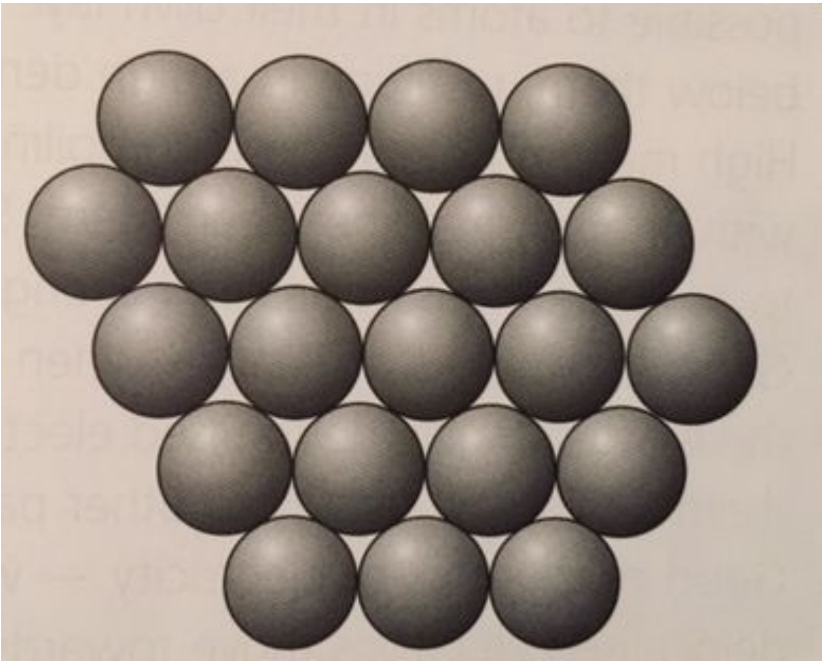
- How is metallic bonding different from covalent bonding?
  - Covalent bonds are localised
  - Metallic bonds are delocalised.





What is a giant metallic structure?

A large three dimensional. structure made of many positive metal ions and delocalised electrons.



Properties of metals

Learning objective

1. State the properties of metals.
2. Explain the metal properties using the knowledge of their structure

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- https://www.youtube.com/watch?v=S08qdOTd0w0

- Watch the video and make notes as you will be answering questions

Questions

- Why is it important that the pot and the coil inside the kettle is made of a metal?
- Other than being good conductors of heat what is another property of metals?
- What is the difference between malleable and ductile?
- What is metallic bonding?
- What is a lattice structure?

Definition

- Metallic bonding is the strong electrostatic attraction between positive metal ion and delocalised electrons.

What are the properties of the metals?

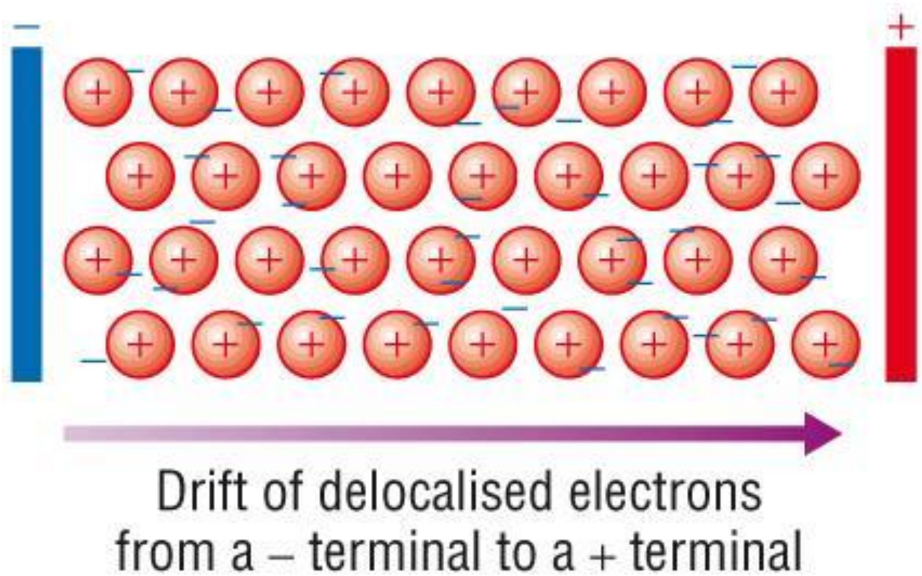
- Have high density
- High melting and boiling points
- Good electrical conductors
- Good conductors of heat
- Malleable
- Ductile

Why do metals have high melting/boiling points?

- The metallic bonds (electrostatic forces) are strong.
- High temperatures are needed to break the metallic bonds.

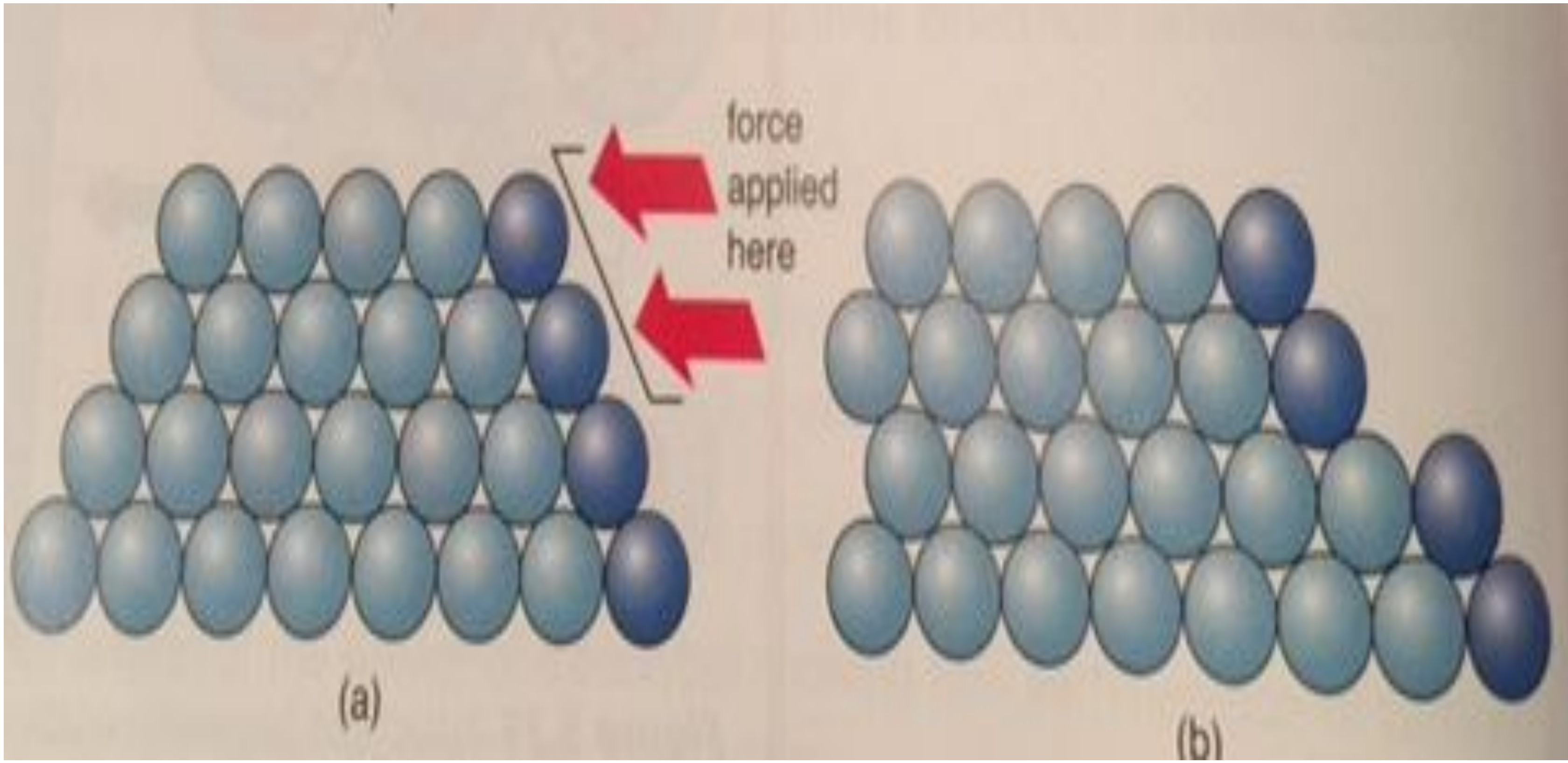
Why do metals conduct electricity?

- The delocalised electrons can move freely anywhere within the metal lattice allowing them to conduce electricity.



Why are metals malleable?

- "Malleable" means that the metal can be hammered and bended to change shape.
- The atoms are able to slide over each other into new positions without breaking the metallic bonds.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(a)
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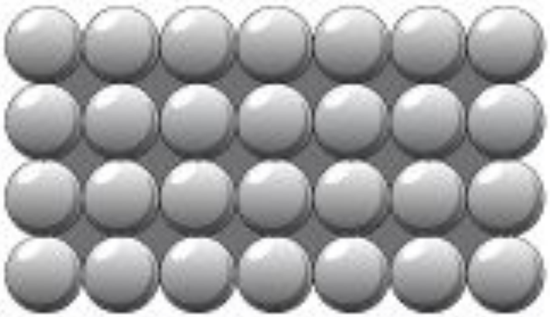
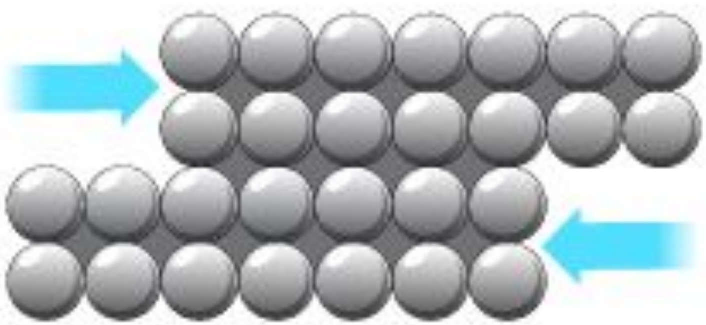
What is an alloy?

- A mixture of metals
- Alloys modify a metal’s properties – how?

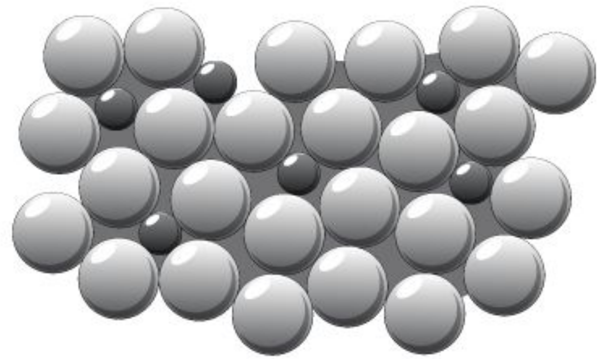
Alloys

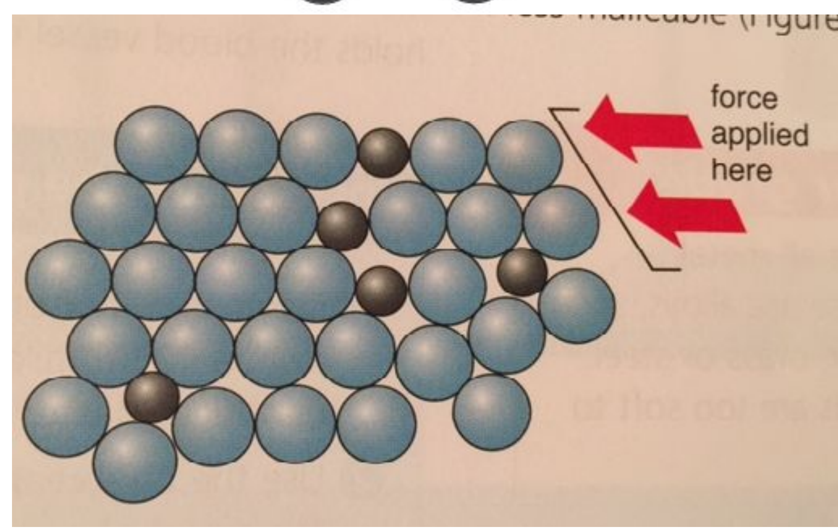
Metal





Alloy





Alloys

- Change the properties of the metals.
- Example: steel
- Iron is mixed with carbon.
- The addition of carbon disrupt the regular arrangement of the iron atoms and lead to the steel being harder and less malleable.

More alloys

- Brass - alloy of zinc and copper
- Bronze- alloy of tin and copper
- Solder – alloy of tin and lead
- Aluminium alloy – aluminium and magnezium or copper

Questions

1) What is meant by metallic bonding?
2) How does a metal conduct electricity?
3) Explain how metallic bonding is different from covalent bonding.